

Frequency modulation

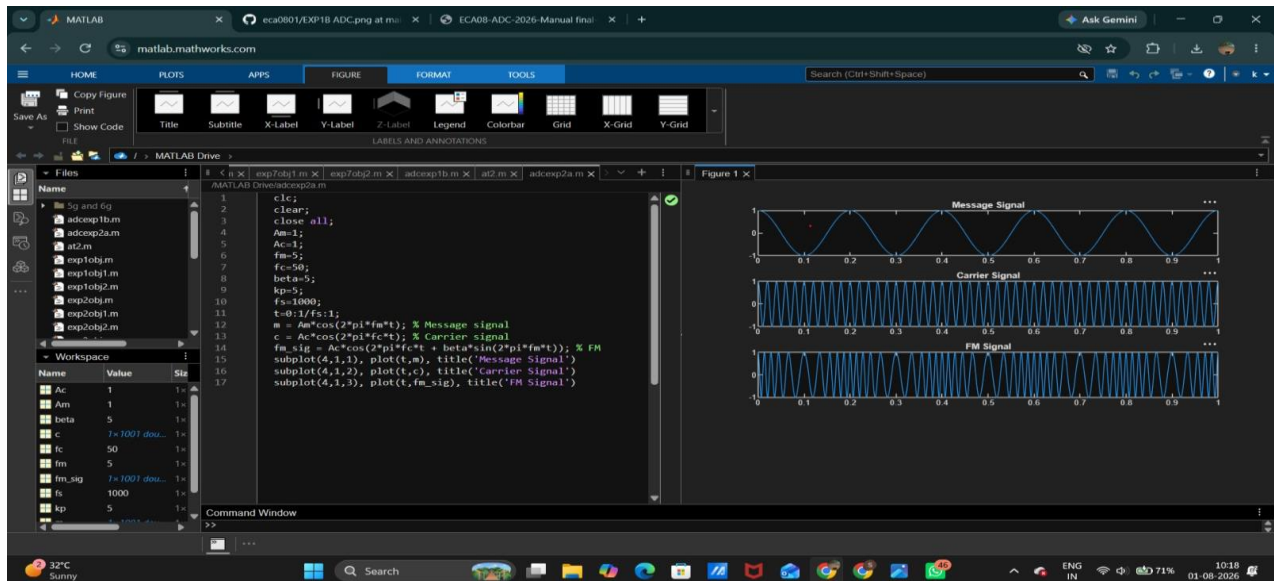
Exp 2a

Objective 1

CODE

```
clc;
clear;
close all;
Am=1;
Ac=1;
fm=5;
fc=50;
beta=2;
kp=5;
fs=1000;
t=0:1/fs:1;
m = Am*cos(2*pi*fm*t);
c = Ac*cos(2*pi*fc*t);
% Message signal
% Carrier signal
fm_sig = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM
subplot(4,1,1), plot(t,m), title('Message Signal')
subplot(4,1,2), plot(t,c), title('Carrier Signal')
subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
```

OUTPUT



Objective 2

CODE

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signals
clc;
clear;
close all;
%% Input Parameters
Am = 21; % Message signal amplitude
fm = 500; % Message frequency (Hz)
Ac = 15; % Carrier amplitude
fc = 1000;
% Carrier frequency (Hz)
kf = 1000; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.02; % Simulation duration (s)
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation Calculation
delta_f = kf * Am;
%% Modulation Index Calculation
beta = delta_f/fm;
%% FM Signal Generation
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Plot Message Signal
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Plot FM Signal
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Spectrum Analysis
N = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal))/N);
f = (-N/2:N/2-1)*(fs/N);
figure;
plot(f,FM_fft,'LineWidth',1.5);
title('FM Signal Spectrum');
```

OUTPUT



Exp 2b

CODE

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
clc;
clear;
close all;
%% Message Signal Parameters
Am = 20;
% Message amplitude (V)
fm = 500;
% Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 12 ;
% Carrier amplitude (V)
fc = 1000;
% Carrier frequency (Hz)
%% FM Parameters
kf = 200;
% Frequency sensitivity (Hz/V)
fs = 10000; % Sampling frequency (Hz)
T = 0.01;
% Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
%% Modulation Index
beta = delta_f/fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (Beta) = %.2f\n',beta);
%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);
%% Verification Check
tolerance = 1e-6;
```

```

if abs(beta - beta_verify) < tolerance
fprintf('\nVerification Successful\n');
else
fprintf('\nVerification Failed\n');
end
%% Plotting
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

```

OUTPUT

